CONNECTING TWO PCS USING CISCO PACKET TRACER

AIM

To design and simulate a simple computer network by connecting two PCs using Cisco Packet Tracer and verify communication between the devices.

OBJECTIVES

- To understand the basics of computer networking.
- To create a simple network topology using Cisco Packet Tracer.
- To connect two PCs using an appropriate network connection.
- To assign IP addresses to both PCs.
- To configure the network correctly.
- To test communication between the two PCs using the ping command.
- To understand how data is transferred between devices in a network.

PROBLEM STATEMENT

In a computer network, devices need to communicate with each other to exchange data and share resources. If two computers are not configured with proper network settings, they cannot communicate successfully.

This case study focuses on creating a simple network consisting of two PCs in Cisco Packet Tracer. The PCs must be connected and configured with suitable IP addresses so that they can communicate with each other. The main challenge is to establish a proper connection and verify that data can be successfully transmitted between the two computers.

SOLUTION

The problem can be solved by creating a simple peer-to-peer network in Cisco Packet Tracer.

Step 1: Add the devices

- Open Cisco Packet Tracer.
- Add two PCs to the workspace.
- Name them PC0 and PC1.

Step 2: Connect the PCs

Connect PC0 and PC1 using an appropriate Ethernet connection. For a direct PC-to-PC connection, Packet Tracer can use a suitable Ethernet cable depending on the interfaces and Packet Tracer version.

Step 3: Configure IP addresses

Assign IP addresses to both PCs in the same network.

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

No default gateway is required for communication between these two PCs on the same local network.

Step 4: Test communication

Open Command Prompt on PC0 and enter**:**

ping 192.168.1.20

If the configuration and physical connection are correct, PC0 should receive replies from PC1.

Similarly, from PC1, enter:

ping 192.168.1.10

Successful replies confirm that the two PCs can communicate.

Expected Result

The network should successfully transmit packets between PC0 and PC1. This demonstrates basic peer-to-peer communication and shows how IP addressing enables devices on the same network to communicate.

CONCLUSION

The two PCs were successfully connected using Cisco Packet Tracer. By assigning IP addresses and using the ping command, communication between the PCs was verified. This demonstrates the basics of network connectivity and IP addressing.